

EXHIBITOR: Siemens Energy STAND: D680

How to make FLNG and Mid-Scale LNG projects more profitable and reliable — with less risk

Sanjeev Daruka, Head of Global LNG, Siemens Energy, with engineering roots and more than 30 years in the world's energy industry, sees tremendous opportunities to improve LNG process design and gas turbine-driven refrigeration compression trains that can support today's dynamic energy transition with an equal emphasis on profitability and local job creation.

In an exclusive interview with Gastech News, he explains why.

What trends are shaping the world's LNG industry today, in general, and compression, in particular?

Rising gas demand and falling domestic gas production in Asia have opened the door for new LNG projects worldwide. We're seeing new LNG projects happening in various countries, such as Argentina and African nations that have large gas reserves but have been unable to capitalize on them so far.

To maximize equity returns and greatly reduce the risks posed by a lack of locally skilled site labor, limited land space, and regulatory compliance, companies are turning more to Floating LNG (FLNG) and modularized Mid-Scale LNG production models than traditional onshore large-scale, stick-built models.

These projects are typically designed to produce 3.5-4.5 MMTA of LNG. And since most of their fabrication is done in shipyards, site risks for onshore assembly and fabrication are minimized.

So, how do Siemens Energy solutions make FLNG and Mid-scale LNG projects more profitable?

With today's CAPEX pressures, LNG project developers are looking to utilize a simplified LNG process solution using single mixed refrigerant (SMR) and operating at the maximum possible capacity to reduce their production cost per ton of LNG. However, that production capacity can be limited by factors such as the gas turbine power, compressor efficiency, and heat



Sanjeev Daruka, Head of Global LNG, Siemens Energy

exchanger capacity. Siemens Energy has worked closely with its clients to develop a solution to address the above limitations and recommends running two parallel compression trains using our lightweight SGT-750 gas turbine and high efficiency compressors in a single SMR LNG train. With total 84 MW power, this configuration allows one SMR LNG train to produce 1.8-2.2 MMTA of LNG, thereby fully utilizing the largest available heat exchanger capacity. Deploying two such SMR trains on a FLNG vessel provides the LNG project investors with the best potential returns possible.

With lost LNG production so costly, what about the reliability and downtime of this solution?

A Siemens Energy SGT-750 gas turbine is ultra-reliable, consistently

achieving 99+% uptime across its installed fleet. Its mean time between overhauls (MTBO) typically exceeds four years — more than industry standards — making it the most reliable solution with minimum downtime.

With two SGT-750 compression trains operating in parallel, even if a unit goes down due to an unscheduled event, the other unit keeps the refrigeration system operating and enables the LNG plant to ramp up to full production immediately after the first unit is back online. Further, the high starting torque capability of SGT-750 enables producers to start from shutdown within minutes, preventing the loss of valuable refrigerant.

How does this solution help with energy transition and environmental goals?

The Siemens Energy SGT-750 gas

turbine is the most environmentally friendly unit in its class. It comes with a guaranteed single-digit NOx emission profile across a wide operating range. In addition, seasonal mapping is not required to maintain its low emission performance, unlike other turbine manufacturers.

What about maintenance and serviceability?

Our lightweight SGT-750 is the industry's most maintenance-friendly industrial gas turbine. The core engine can be exchanged with a replacement within 24 hours.

What's more, the original engine can be overhauled in-situ or in-country without a need to send back to the factory. This feature saves our customers the significant cost, downtime, and administrative burden that are due to international shipping with associated customs issues and delays. At the same time, it also helps create local jobs, making a direct impact on a country's economy.

In addition, with our compressor operating at 6100 rpm, the bundle maintenance weight of our solution is less than 35 tons. That is less than half the weight of our competitors' solutions, which typically weigh 80-100 tons. This eliminates the need for a massive crane for maintenance, reduces CAPEX, and improves safety aspects.

Finally, the entire turbo-compression package operates on one common oil system, eliminating the need for separate oil systems, additional components, and reducing maintenance and CAPEX.